[12/25/2020 14:21] >[NPpTHtxvpIg]

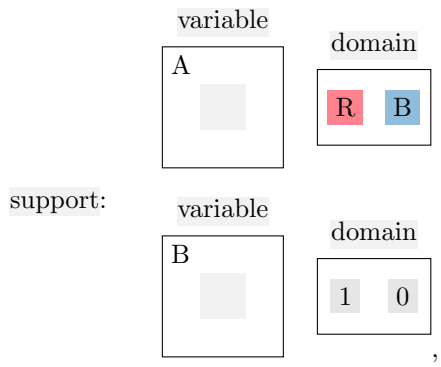
[35] /T/gr/T/pr

Probability Theory

[35.1] >

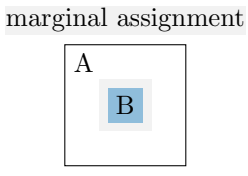
Physical intuition.

Suppose that we have a set of "variables," each with their own finite "domains", which we'll collectively refer to as the "support": [35.2] >



,

where a "marginal assignment" consists of associating some (but not all) of the variables of the support with something from their domains: [35.3] >



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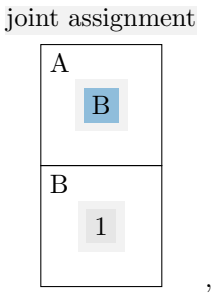
a "joint assignment" ("assignment" for short) consists of associating each variable of the support with something from its domain: [35.4] >



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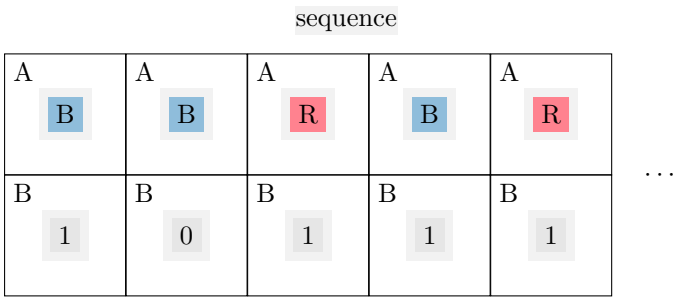
a "sequence" consists of an infinite, ordered list of such joint assignments: [35.5] >



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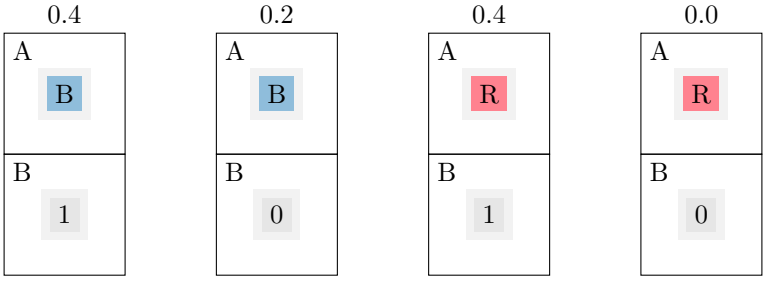
and a "trial" denotes a particular item (i.e. an indexed joint assignment) of this sequence.

Now, suppose something in particular about this sequence: if a joint assignment occurs at any trial, it will occur in infinitely many trials, and, additionally, as we the traverse sequence in order, the ratio of occurrences of the joint assignment to trials so far becomes arbitrarily close to some positive number. Otherwise, if an joint assignment never occurs, the ratio remains at zero. Let's call this ratio the "sequence probability" of the joint assignment. Let's refer to such a sequence as a probability sequence. [35.6] >

For example, if the sequence above repeated the first five trials, it would be a probability sequence defining the sequence probabilities: [35.7] >

sequence probabilities



Let a "distribution" be a set of probability sequences with the same sequence probability for every assignment, which, in the context of a distribution, we will simply refer to as the "joint probability" ("probability" for short) of that assignment. [35.8] >

Notation.

[Define]$^{36}_{\mathrm{D}}$

$$\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}} \coloneqq \text{an arbitrary set of variable(s) with } N_{\mathbf{X}} \geq 1 \text{ and domains } \mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_{N_{\mathbf{X}}},$$

and we define $\mathrm{Y}_1, \mathrm{Y}_2, \ldots, \mathrm{Y}_{N_{\mathbf{Y}}}/\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_{N_{\mathbf{Y}}}/N_{\mathbf{Y}}$, $\mathrm{Z}_1, \mathrm{Z}_2, \ldots, \mathrm{Z}_{N_{\mathbf{Z}}}/\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_{N_{\mathbf{Z}}}/N_{\mathbf{Z}}$, and $\mathrm{W}_1, \mathrm{W}_2, \ldots, \mathrm{W}_{N_{\mathbf{W}}}/\mathbf{W}_1, \mathbf{W}_2, \ldots, \mathbf{W}_{N_{\mathbf{W}}}/N_{\mathbf{W}}$ similarly.

Define

$$?_1, ?_2, \ldots, ?_{N_{\mathbf{?}}} \coloneqq \text{an arbitrary set of variable(s) with } N_{\mathbf{?}} \geq 0 \text{ and domains } \mathbf{?}_1, \mathbf{?}_2, \ldots, \mathbf{?}_{N_{\mathbf{?}}}.$$

Define $?'_1, ?'_2, \ldots, ?'_{N_{\mathbf{?}'}}/\mathbf{?}'_1, \mathbf{?}'_2, \ldots, \mathbf{?}'_{N_{\mathbf{?}'}}/N_{\mathbf{?}'}$ similarly.

[Define]$^{102}_{\mathrm{D}}$

$$\mathrm{X} \coloneqq [\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}}]$$

with

$$\mathbf{X} \coloneqq [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_{N_{\mathbf{X}}}].$$

Define $\mathrm{Y}/\mathbf{Y}$, $\mathrm{Z}/\mathbf{Z}$, $\mathrm{W}/\mathbf{W}$, $?/\mathbf{?}$ and $?'/\mathbf{?}'$ similarly.

Note that $\|\mathrm{X}\| = N_{\mathbf{X}} \geq 1$, $\|\mathrm{Y}\| = N_{\mathbf{Y}} \geq 1$, $\|\mathrm{Z}\| = N_{\mathbf{Z}} \geq 1$, $\|\mathrm{W}\| = N_{\mathbf{W}} \geq 1$, $\|?\| = N_{\mathbf{?}} \geq 0$ and $\|?'\| = N_{\mathbf{?}'} \geq 0$.

[Let the notation]$^{103}_{\mathrm{D}}$

$$(\mathrm{X}, \mathbf{X}) \text{ indicate a “matching” between variables } \mathrm{X} \text{ and domains } \mathbf{X},$$

which expands as

$$(\mathrm{X}_1, \mathbf{X}_1), (\mathrm{X}_2, \mathbf{X}_2), \ldots, (\mathrm{X}_{N_{\mathbf{X}}}, \mathbf{X}_{N_{\mathbf{X}}}).$$

[Let the notation]$^{37}_{\mathrm{D}}$

$$\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\} \text{ establish a support consisting of variables } \mathrm{X} \text{ with domains } \mathbf{X}.$$

[Let]$^{57}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ [or, $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$] and all $\mathrm{x} = [\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_{N_{\mathbf{X}}}] \in \mathbf{X}^{\times}$,

$$\mathrm{X} = \mathrm{x}$$

denote a joint assignment (assignment for short) [or, marginal assignment], which expands as

$$\mathrm{X}_1 = \mathrm{x}_1, \mathrm{X}_2 = \mathrm{x}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}} = \mathrm{x}_{N_{\mathbf{X}}}.$$

If a joint assignment [or, marginal assignment] was previously specified, we may later drop “X =”, for “implied assignment”:

$$\mathrm{x},$$

which expands as

$$\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_{N_{\mathbf{X}}}.$$

If a joint assignment [or, marginal assignment] was never specified, we assume an arbitrary $\mathrm{x} \in \mathbf{X}^{\times}$ for “implied generality”:

$$\mathrm{X},$$

which expands as

$$\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}}.$$

[Let]$^{38}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQ} \text{ denote the set of all sequences.}$$

[Use]$^{63}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $1 \leq t < \infty$, $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$,

$$I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) \coloneqq \begin{cases} 1 & \text{trial } t \text{ of sequence sq has assignment } \mathrm{X} = \mathrm{x} \\ 0 & \text{otherwise} \end{cases}$$

to describe a sequence $\mathrm{sq} \in \mathbf{SQ}$, where

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) = 1.$$

[Let]$^{64}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $1 \leq t < \infty$ and $\mathrm{sq} \in \mathbf{SQ}$,

$$I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) \coloneqq \sum_{? \in \mathbf{?}^{\times}} I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?\right).$$

[Define]$^{40}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, all $\mathrm{sq} \in \mathbf{SQ}$, and all $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) \coloneqq \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right)}{t} \right);$$

that is, the sequence probability of assignment $\mathrm{X} = \mathrm{x}$ in sequence sq.

[Define]$^{66}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) \coloneqq \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right)}{t} \right);$$

that is, the marginal sequence probability of marginal assignment $\mathrm{X} = \mathrm{x}$ in sequence sq.

[Define]$^{73}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) \coloneqq \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right)}{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{Y} = \mathrm{y}\right)} \right);$$

that is, the conditional sequence probability of marginal assignment $\mathrm{X} = \mathrm{x}$ in sequence sq given marginal assignment $\mathrm{Y} = \mathrm{y}$.

[Let]$^{41}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQP} = \{\mathrm{sq} \in \mathbf{SQ} \mid \forall \mathrm{x} \in \mathbf{X}^{\times}, (\exists\, 1 \leq t < \infty : I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) = 1) \implies \Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) > 0\},$$

that is, the set of all probability sequences.

Propositions.

[It can be shown]$^{69}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ and all $\mathrm{x} \in \mathbf{X}^{\times}$ and $\mathrm{sqp} \in \mathbf{SQP}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \neq [\mathrm{null}].$$

[It can be shown]$^{71}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = \sum_{? \in \mathbf{?}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, ? = ?).$$

[It can be shown]$^{90}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$, $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

[It can be shown]$^{81}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{x} \in \mathbf{X}^{\times}$ and $\mathrm{sqp} \in \mathbf{SQP}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \neq [\mathrm{null}].$$

[It can be shown]$^{91}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = 0 \implies \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = 0.$$

[It can be shown]$^{92}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) > 0 \implies \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) > 0.$$

[It can be shown]$^{98}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $1 \leq t < \infty$, $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$,

$$I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = \begin{cases} 1 & \exists\, ? \in \mathbf{?}^{\times} : I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?) = 1 \\ 0 & \text{otherwise} \end{cases}$$

where

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = 1.$$

[It can be shown]$^{97}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $1 \leq t < \infty$ and $\mathrm{sqp} \in \mathbf{SQP}$,

$$\exists\, 1 \leq t < \infty : I(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}) = 1 \iff \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) > 0.$$

[It can be shown]$^{74}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})}.$$

[It can be shown]$^{89}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0 \iff \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}].$$

[It can be shown][93] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \neq [\mathrm{null}] \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}].$$

[It can be shown][94] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \times \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) = \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

[It can be shown][95] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, if $\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 1 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 1.$$

[It can be shown][96] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, if $\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 0.$$

[It can be shown][116] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Z}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) = \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}).$$

[It can be shown][118] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}.$$

[It can be shown][119] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) = 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) = 0.$$

[It can be shown][120] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) > 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) > 0.$$

[It can be shown][121] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$,

$$\begin{aligned} \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) > 0 \implies & \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \times \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) \\ & = \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}). \end{aligned}$$

[It can be shown][128] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x})}{t} \right) \neq [\mathrm{null}]$$

[It can be shown][127] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$,

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) = 1.$$

[It can be shown][129] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$,

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) = 1.$$

[It can be shown][130] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, and all $\mathrm{y} \in \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0 \implies \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 1.$$

[It can be shown][131] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?}), (?', \mathbf{?}')\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, ? = ?) > 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) > 0.$$

Notation.

[Define]$^{42}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{D} = \{\emptyset \subset \mathrm{D} \subseteq \mathbf{SQP} \mid \forall\, \mathrm{d}, \mathrm{d}' \in \mathrm{D}^2, \forall\, \mathrm{x} \in \mathbf{X}^{\times}, \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right)\}$$

that is, the set of all distributions.

[Define]$^{43}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, $\mathrm{D} \in \mathbf{D}$ and all $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr(\mathrm{X} = \mathrm{x})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr(\mathrm{X} = \mathrm{x}) = \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}),$$

that is, a joint probability.

[Define]$^{46}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr(\mathrm{X} = \mathrm{x})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr(\mathrm{X} = \mathrm{x}) = \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}),$$

that is, a marginal probability.

[Define]$^{48}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y});$$

that is, a conditional probability.

Notation.

[Define]$^{60}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $f : \mathbf{X} \to \mathbb{R}$,

$$\mathbb{E}\left[\mathrm{sq}; f\right] = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x})}{t} \right).$$

This is verbally referred to as the "sequence expectation" of f under sq.

[Define]$^{61}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$, $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$ and $\mathrm{y} \in \mathbf{Y}^{\times}$,

$$\mathbb{E}\left[\mathrm{sq}; f \mid \mathrm{Y} = \mathrm{y}\right] = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y})}{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{Y} = \mathrm{y}\right)} \right).$$

This is verbally referred to as the "conditional sequence expectation" of f under sq given $\mathrm{Y} = \mathrm{y}$.

Propositions.

[It can be shown]$^{83}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $f : \mathbf{X} \to \mathbb{R}$,

$$\mathbb{E}\left[\mathrm{sq}; f\right] = \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x}).$$

[It can be shown]$^{72}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$ and all $\mathrm{sqp} \in \mathbf{SQP}$, $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$, and $\mathrm{y} \in \mathbf{Y}^{\times}$,

$$\mathbb{E}\left[\mathrm{sqp}; f \mid \mathrm{Y} = \mathrm{y}\right] = \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y}).$$

[It can be shown]$^{100}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$,

$$\mathbb{E}\left[\mathrm{sqp}; f\right] = \sum_{\mathrm{y} \in \mathbf{Y}^{\times} : \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0} \mathbb{E}\left[\mathrm{sqp}; f \mid \mathrm{Y} = \mathrm{y}\right] \times \Pr\left(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}\right).$$

Notation.

[Define]$^{58}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, $\mathrm{D} \in \mathbf{D}$, and all $f : \mathbf{X} \to \mathbb{R}$,

$$\mathbb{E}\left[f\right]$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \mathbb{E}\left[f\right] = \mathbb{E}\left[\mathrm{d}; f\right].$$

This is verbally referred to as the "expectation" of f under D.

[Define]$^{67}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, $\mathrm{D} \in \mathbf{D}$, and all $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$ and $\mathrm{y} \in \mathbf{Y}^{\times}$,

$$\mathbb{E}\left[f \mid \mathrm{Y} = \mathrm{y}\right]$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \mathbb{E}\left[f \mid \mathrm{Y} = \mathrm{y}\right] = \mathbb{E}\left[\mathrm{d}; f \mid \mathrm{Y} = \mathrm{y}\right].$$

This is verbally referred to as the "conditional expectation" of f under D given $\mathrm{Y} = \mathrm{y}$.

Notation.

[Define]$^{113}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}, \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}] \implies \Pr(\mathrm{x} \mid \mathrm{z}, \mathrm{y}) = \Pr(\mathrm{x} \mid \mathrm{z}),$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \iff \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \emptyset \iff \forall \mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}, \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}] \implies \Pr(\mathrm{x} \mid \mathrm{y}) = \Pr(\mathrm{x}).$$

Propositions.

[It can be shown]$^{115}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) = \Pr(\mathrm{x} \mid \mathrm{z}) \times \Pr(\mathrm{y} \mid \mathrm{z}),$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \iff \forall \mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}, \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = \Pr(\mathrm{x}) \times \Pr(\mathrm{y}).$$

[It can be shown]$^{122}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\begin{aligned} \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff & \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \\ & \Pr(\mathrm{Z} = \mathrm{z}) > 0 \implies \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = \Pr(\mathrm{x} \mid \mathrm{z}) \times \Pr(\mathrm{y}, \mathrm{z}). \end{aligned}$$

[It can be shown]$^{114}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff \mathrm{Y} \perp\!\!\!\perp \mathrm{X} \mid \mathrm{Z},$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \iff \mathrm{Y} \perp\!\!\!\perp \mathrm{X}.$$

[It can be shown]$^{123}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (\mathrm{W}, \mathbf{W}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{W} \mid \mathrm{Z} \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z},\ \mathrm{X} \perp\!\!\!\perp \mathrm{W} \mid \mathrm{Z},$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z} \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y},\ \mathrm{X} \perp\!\!\!\perp \mathrm{Z}.$$

[It can be shown]$^{124}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (\mathrm{W}, \mathbf{W}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{W} \mid \mathrm{Z} \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z}, \mathrm{W},$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z} \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z}.$$

[It can be shown]$^{125}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (\mathrm{W}, \mathbf{W}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z},\ \ \mathrm{X} \perp\!\!\!\perp \mathrm{W} \mid \mathrm{Z}, \mathrm{Y}, \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{W} \mid \mathrm{Z},$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y},\ \ \mathrm{X} \perp\!\!\!\perp \mathrm{Z} \mid \mathrm{Y}, \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z}.$$

[It can be shown]$^{126}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (\mathrm{W}, \mathbf{W}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z}, \mathrm{W},\ \ \mathrm{X} \perp\!\!\!\perp \mathrm{W} \mid \mathrm{Z}, \mathrm{Y},\ \ \forall \mathrm{y}, \mathrm{z}, \mathrm{w} \in \mathbf{Y}^{\times} \times \mathbf{Z}^{\times} \times \mathbf{W}^{\times}, \Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}, \mathrm{W} = \mathrm{w}) > 0$$

$\implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{W} \mid \mathrm{Z}$, and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z},\ \ \mathrm{X} \perp\!\!\!\perp \mathrm{Z} \mid \mathrm{Y},\ \ \forall \mathrm{y}, \mathrm{z} \in \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0 \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z}.$$

Notation.

[Define]$^{132}_{\mathtt{D}}$, for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, and all $\mathrm{E} \subseteq \mathrm{V}^2$, with $\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$,

$$\mathrm{IMAP}\,(\mathrm{G}) \iff \forall\, \mathrm{C} \subseteq \mathrm{V},\ \emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V} : \mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}),\ \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \impliedby (\mathrm{G};\ \mathrm{A} \not\to \mathrm{B} \setminus \mathrm{C}).$$

[Define]$^{133}_{\mathtt{D}}$, for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, and all $\mathrm{E} \subseteq \mathrm{V}^2$, with $\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$,

$$\mathrm{DMAP}\,(\mathrm{G}) \iff \forall\, \mathrm{C} \subseteq \mathrm{V},\ \emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V} : \mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}),\ \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \implies (\mathrm{G};\ \mathrm{A} \not\to \mathrm{B} \setminus \mathrm{C}).$$

[Define]$^{134}_{\mathtt{D}}$, for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, and all $\mathrm{E} \subseteq \mathrm{V}^2$, with $\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$,

$$\mathrm{PMAP}\,(\mathrm{G}) \iff \mathrm{IMAP}\,(\mathrm{G}),\ \mathrm{DMAP}\,(\mathrm{G}).$$

Propositions.

[It can be shown][136]$_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, if for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B}, \mathrm{S} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},\ \mathrm{A} \perp\!\!\!\perp \mathrm{S} \mid \mathrm{C}, \mathrm{B} \implies \mathrm{A} \perp\!\!\!\perp \mathrm{B}, \mathrm{S} \mid \mathrm{C},$$

for all $\mathrm{C}, \mathrm{S} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \implies \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},$$

and for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \implies \forall \mathrm{H} \in \mathrm{V} \setminus (\mathrm{A}, \mathrm{B}, \mathrm{C}),\ \mathrm{A} \perp\!\!\!\perp [\mathrm{H}] \mid \mathrm{C} \text{ or } [\mathrm{H}] \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}$$

then $\exists \mathrm{E} \subseteq \mathrm{V}^2 : \mathrm{PMAP}\,([\mathrm{V}, \mathrm{E}])$.

Topics.

[Reinforcement Learning Theory]$^{2}_{\mathrm{T}}$